

Question	Answer	Marks
3(a)(i)	force / (cross-sectional) area	B1
3(a)(ii)	extension / original length	B1
3(b)(i)	measure / determine / find diameter	B1
	using a micrometer / <u>digital</u> calipers	B1
	several measurements in different places / along the wire / around the circumference (and average them)	B1
3(b)(ii)	$E = \sigma / \epsilon$ or $E = FL / Ax$ or $E = \text{gradient} \times (L / A)$ $E = (4 \times 2.5) / (0.8 \times 10^{-3}) \times (9.4 \times 10^{-8})$	C1
	$= 1.3 \times 10^{11} \text{ Pa}$	A1

Question	Answer	Marks
3(b)(iii)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$ $E = \frac{1}{2} \times (2+4) \times 0.4 \times 10^{-3}$ or $E = (\frac{1}{2} \times 4 \times 0.8 \times 10^{-3}) - (\frac{1}{2} \times 2 \times 0.4 \times 10^{-3})$ or $E = [\frac{1}{2} \times 5000 \times (0.8 \times 10^{-3})^2] - [\frac{1}{2} \times 5000 \times (0.4 \times 10^{-3})^2]$	C1
	$E = 1.2 \times 10^{-3} \text{ J}$	A1
3(c)	straight line from the origin and above the original line	M1
	straight line passes through (0.80, 8.0)	A1

Question	Answer	Marks
4(a)	(two) waves (travelling at same speed) in opposite directions overlap	B1